Exercise 2

1. Boost Converter

A boost converter is a DC/DC power converter which steps up voltage from its input (source) to its output (load). In continuous conduction mode (current through the inductor never falls to zero), the theoretical transfer function of the boost converter is:

$$\frac{V_{out}}{V_{in}} = \frac{1}{1-D}$$

where *D* is the duty cycle.

1) In this exercise, the converter is feeding an RC load from a 24 V source and the control PWM signal is set to 20 kHz and duty cycle is 50%.
2) Observe and record the voltage and current waveform on inductor.
3) Verify the volt-second balance in steady state.
4) Run the simulation and observe waveforms on Scope. Verify that the mean value of the load voltage (V_{out}) is very close to the theoretical value of: $V_{out} = 24/(1-0.5) = 48\text{V}$.
5) If the source of the value change to 20V and the output voltage as same as above, adjust the duty cycle of PWM and observe the waveforms on scope.

2. Transient Response of a Buck–Boost Converter

In addition to steady-state conversion characteristics, it is often of interest to investigate converter transient responses. For example, in voltage regulator designs, it is necessary to verify whether the output voltage remains within specified limits when the load current takes a step change. As another example, during a start-up transient when the converter is powered up, converter components can be exposed to significantly higher stresses than in steady state. It is of interest to verify that component stresses are within specifications or to make design modifications to reduce the stresses. In this experiment, transient simulations can be used to test for converter responses.

Transient simulations can be performed on the converter switching circuit model or on the converter averaged circuit model. As an example, let us apply these two approaches to investigate a start-up transient response of the buck-boost converter shown in below.

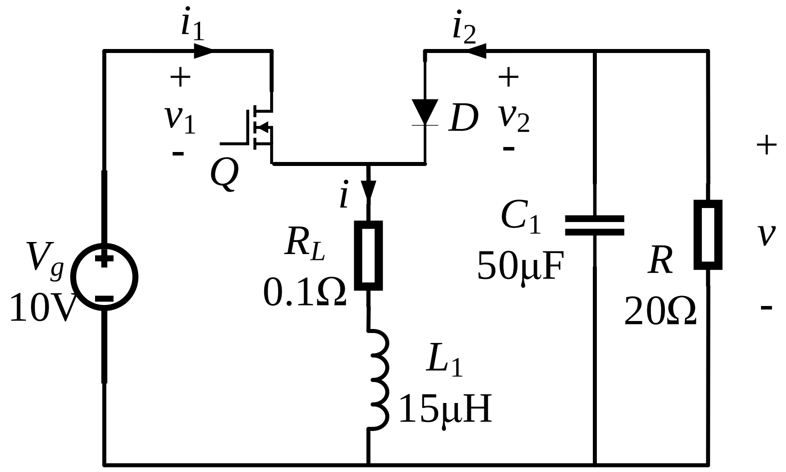

Fig 1

The inductor winding resistance R_L is included to model the inductor copper losses. The MOSFET is controlled by a pulsating voltage source v_1. The switch MOSFET switch on-resistance $R_{on} = 50\text{m}\Omega$. The switch is on when the controlling voltage v_1 is greater than $v_{on} > 0$, and off when the controlling voltage v_1 is less than $v_{off} \leq 0$. The pulsating source v_1 has the pulse amplitude equal to 10V, the period is $T_s = 1/f_s = 10\mu\text{s}$. The diode on-resistance $R_{on} = 5\text{m}\Omega$, and the voltage drop $V_D = 0.7\text{V}$. Conduction losses are modeled in a simple manner, and details of complex device behavior during switching transitions are neglected.

1) Observe inductor current and capacitor voltage waveforms during the start-up transient. Switching ripples can be observed in the waveforms obtained by simulation of the switching circuit model.
2) Based on the results above, we can see that converter components are exposed to significantly higher current stresses during the start-up transient than during steady state operation. The problem of excessive stresses in the start-up transient is quite typical for switching power converters. How could you reduce the start-up transient stresses from start-up to the steady-state value?

3) Compute the controlling frequency*

A 100μH inductance is used as the CCM inductance of a buck/boost converter with an output voltage of 144VDC and an expected input voltage variation of 120 to 162 VDC. The expected output current variation is 5 to 10 amperes. What is the minimum PWM switching frequency that ensures CCM operation?

*: You can not simulate this question if you are not interested or have no time.

Marking scheme

Each lab session will be assessed and will count towards the overall mark for the practical component of this course. The lab demonstrator will check your measurements and the quality of presentation of data in the lab book and plots.

Laboratory report (100/100 marks)

- Objective & background (total mark - 30/**100**)
- Calculations, graphs, and results (total mark - 40/**100**)
- Discussion of results and conclusions (total mark - 30/**100**)